

# Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks

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**Abstract**—Static random access memory (SRAM) based compute-in-Memory(CIM) arrays has emerged as an effective approach to mitigate the memory bandwidth bottleneck in conventional von Neumann architectures. This study presents a 6T CMOS SRAM-based in-memory computing architecture designed in 45 nm technology for carrying out Boolean logic operations directly within the memory array. Circuit-level functionality is validated by transient simulations, and Monte Carlo analysis evaluates process-associated variations in propagation delay of logic operations. The delay distributions are used to determine timing-induced logic errors as a function of clock margin. A hardware-aware binary neural network (BNN) framework is developed to evaluate the system-level impact of timing-induced errors, incorporating circuit-level XNOR error probabilities into the inference process for MNIST digit classification. Experimental results demonstrate that reducing the XNOR error probability from 0.095 to 0.0025 improves classification accuracy from 58.27% to 89.34%. Additionally, Monte Carlo energy analysis indicates a mean energy consumption of approximately 3.17 pJ per CIM operation, corresponding to an estimated inference energy of about 12.27  $\mu$ J for the implemented BNN model. The proposed framework connects circuit-level time variation of SRAM-based Boolean CIM operations to the inference accuracy of BNN through Circuit-to-application evaluation.

**Index Terms**—Compute-in-Memory (CIM), SRAM, Binary Neural Networks (BNN), Timing Variability, Monte Carlo Analysis, Delay, Inference accuracy

## I. INTRODUCTION

The rapid development of data-intensive applications such as artificial intelligence, neural networks, machine learning, and edge computing has placed significant pressure on conventional computing architectures. In traditional von Neumann systems, memory and processing units are physically separated, requiring frequent data transfers between them. This

separation leads to substantial latency and energy overhead, commonly referred to as the von Neumann bottleneck, which has become a critical challenge for modern computing systems [1], [2]. To mitigate this limitation, compute-in-memory (CIM) and in-memory computing (IMC) paradigms have been introduced that enable computation directly within memory arrays. CIM architectures can significantly improve energy efficiency and throughput by reducing unnecessary data movement between memory and processing units [1], [3], [4].

Among various memory technologies, static random access memory (SRAM) based CIM has achieved significant recognition due to its compatibility with standard CMOS processes, fast access time, and support for large-scale parallel operations [5]–[10].

Most existing SRAM-based CIM implementations target conventional convolutional neural networks (CNNs) and focus on optimizing multiply-accumulate (MAC) operations in memory arrays, which require additional peripheral circuitry and can be sensitive to device variations. In contrast, digital Boolean operations provide a simpler and more robust alternative for in-memory computing, and binary neural networks (BNNs) naturally align with this paradigm by replacing multiplications with Boolean operations such as XNOR followed by accumulation. Recent studies have demonstrated SRAM-based compute-in-memory (CIM) architectures capable of performing Boolean logic and neural-network inference directly within memory arrays [6]. In particular, 6T SRAM-based designs have shown support for multiple Boolean operations without complex additional circuitry [1], [2], [10].

However, circuit-level nonidealities such as process variations and timing uncertainties introduce propagation-delay variations in logic gates. As SRAM technologies scale to advanced nodes, device variability and parasitic effects further

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impact circuit reliability [11]. These nonidealities can propagate through the computation pipeline and degrade neural network inference accuracy, motivating cross-layer evaluation frameworks that connect circuit behavior with application-level performance [12]. To the best of our knowledge, prior work has not reported a circuit-to-application evaluation framework that links timing variations in SRAM-based Boolean CIM operations to BNN inference accuracy. The main contributions of this work are summarized as follows:

- i. Monte Carlo simulations are performed to characterize the propagation-delay variations and evaluate the timing reliability of the implemented Boolean gates.
- ii. A timing error model is developed to estimate logic error probability as a function of clock margin based on propagation delay distributions.
- iii. A hardware-aware Binary Neural Network evaluation framework is implemented by injecting timing-induced XNOR errors into MNIST inference.
- iv. A relationship between hardware timing reliability and neural network inference accuracy is demonstrated, where increasing the clock margin from  $k = 1.005$  to  $k = 1.007$  reduces XNOR error probability from 0.095 to 0.0025 and improves classification accuracy from 58.27% to 89.34%.

## II. PROPOSED CIM SRAM ARCHITECTURE

This section presents the proposed SRAM-based compute-in-memory (CIM) architecture that implements Boolean logic operations directly within the memory array using standard 6T SRAM cells. SRAM-based CIM approaches contribute to system efficiency by executing computations directly within the memory array, hence minimizing data transfers between memory and processing units. Beyond logic-level operations, recent SRAM-based CIM architectures have also demonstrated support for arithmetic computation. Wang *et al.* proposed a compute SRAM architecture capable of performing logic and arithmetic operations within the memory array using bit-serial computation [9]. Similarly, Lee *et al.* introduced a bit-parallel CIM architecture based on conventional 6T SRAM cells that supports reconfigurable precision arithmetic while maintaining compatibility with standard SRAM designs [10].

### A. 6T SRAM based CIM Structure

The proposed architecture shown in Fig. 1(a) utilizes two conventional 6T SRAM cells connected through shared bitline structures to perform Boolean computations directly inside the memory array. Each SRAM cell stores a binary operand, and logic operations are realized by simultaneously activating the corresponding wordlines (WL1, WL2) of the associated cells. When both cells are accessed simultaneously, the stored values interact through the bitlines (BL, BLB), producing voltage variations that facilitate the Boolean operations. Compared to conventional memory, where only one wordline is activated per read, CIM activates multiple wordlines simultaneously which allows bitline discharge patterns to encode Boolean operations

such as AND, OR, XOR, and XNOR depending on stored data and sensing method.

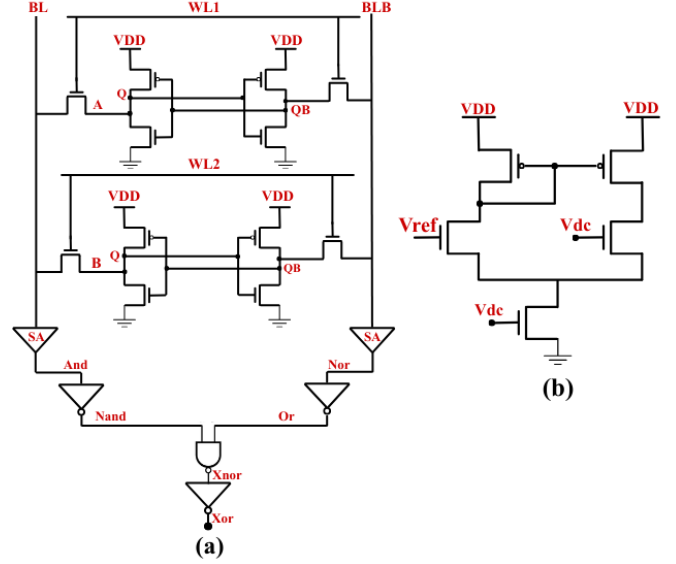


Fig. 1: (a) Proposed CIM enabled 6T SRAM design (b) Differential Sense Amplifier Design

### B. Differential Sense Amplifier for CIM Readout

During CIM operation, the simultaneous activation of multiple wordlines produces small variations in the bitline voltage depending on the stored data values. The differential sense amplifier shown in Fig. 1(b) detects these voltage differences and generates a digital output corresponding to the evaluated Boolean function. Similar sensing approaches have been used in prior SRAM-based CIM architectures [1], [9].

### C. Supported Boolean Operations

The proposed SRAM CIM architecture supports a range of Boolean logic operations including AND, OR, NAND, NOR, XOR, and XNOR. Fig. 2 represents the transient response of all the individual gates. The XNOR function is particularly significant for machine learning applications, among these operations. In binary neural networks (BNNs), XNOR operations followed by population count accumulation can substitute conventional neural network multiplication. Hence, efficient implementation of XNOR operations within memory arrays can significantly accelerate neural network inference.

## III. CIRCUIT IMPLEMENTATION AND SIMULATION

The proposed SRAM-based CIM circuit is designed and simulated in Cadence Virtuoso using a 45 nm CMOS technology library. Monte Carlo simulations are performed in Cadence Maestro to analyze process-variation effects on timing. For each implemented Boolean gate, 1200 samples are generated to obtain propagation delay distributions under process variations as shown in Fig. 4. From these distributions, the mean and standard deviation of propagation delay are

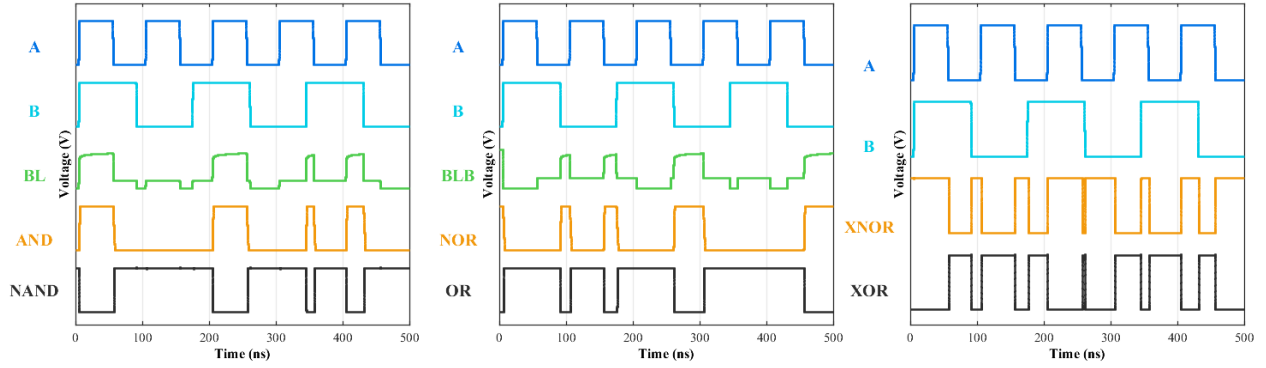


Fig. 2: Transient Response of Individual Gates

extracted to build a probabilistic timing-error model used in the reliability analysis.

The propagation delay distributions obtained from Monte Carlo simulations are used to evaluate the timing reliability of the proposed SRAM-based compute-in-memory (CIM) architecture. Under process variations, transistor parameters such as threshold voltage and carrier mobility vary across devices, resulting in statistical variations in circuit delay that may lead to timing violations when operating with limited clock margins. Let  $T_{delay}$  denote the propagation delay of a logic operation and  $T_{clk}$  represent the clock period. A timing error occurs when

$$T_{delay} > T_{clk} \quad (1)$$

Thus, the probability of timing error can be expressed as

$$P_{error} = P(T_{delay} > T_{clk}) \quad (2)$$

From the Cadence Monte Carlo simulations, the propagation delay distribution of all the gates is obtained using 1200 random samples. To evaluate timing reliability, the clock period is expressed as a scaled multiple ( $k$ ) of the mean delay ( $\mu_{delay}$ ),

$$T_{clk} = k \cdot \mu_{delay} \quad (3)$$

#### IV. TIMING RELIABILITY ANALYSIS

The probability of timing error for each Boolean operation is evaluated by sweeping the clock margin factor  $k$  from 1.005 to 1.007 and computing the number of Monte Carlo delay samples that violate the clock constraint. Fig. 3 illustrates the relationship between the clock scaling factor and the resulting error probability for all logic gates implemented, including AND, OR, NAND, NOR, XOR, and XNOR.

The results indicate that different Boolean operations exhibit significantly different timing sensitivities under process variation. The OR and NOR gates demonstrate the highest timing error probability. This occurs because the OR/NOR function in SRAM CIM frequently relies on bitline discharge via stacked transistors leading to larger propagation delay which, in turn, results in higher timing error probability. However, XOR and XNOR operations exhibit substantially lower error rates

and improved timing robustness. This is because the bitline differential voltage tends to be more distinct.

#### V. BNN HARDWARE-AWARE EVALUATION

A hardware-aware binary neural network (BNN) model is developed to evaluate the system-level impact of timing-induced logic errors. In BNN inference, both weights and activations are represented using binary values, allowing multiplication to be replaced by XNOR operations. Therefore, the reliability of the XNOR logic gate directly determines the accuracy of neural network computations.

This work incorporates the timing error probability derived from the Monte Carlo delay analysis of the XNOR gate into the BNN inference process. *Training Hyperparameters:* Optimizer = Adam, learning rate =  $1 \times 10^{-3}$ , loss function = CrossEntropyLoss, epochs = 10, training batch size = 64,

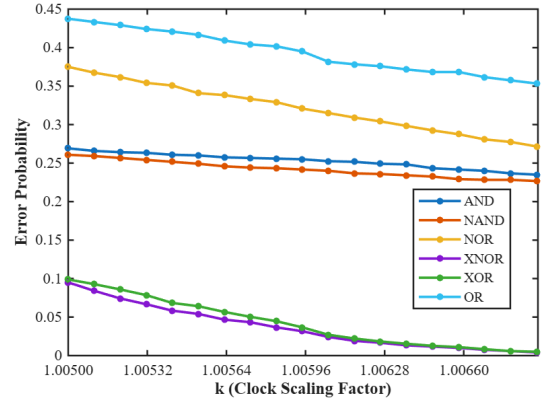


Fig. 3: Timing-induced error probability of the logic gates as a function of clock margin.

TABLE I: Impact of timing-induced XNOR errors on MNIST BNN accuracy

| k      | $T_{clk}$ (ns) | $P_{error}$ | MNIST Accuracy |
|--------|----------------|-------------|----------------|
| 1.0050 | 108.657        | 0.095       | 58.27%         |
| 1.0055 | 108.714        | 0.054       | 70.84%         |
| 1.0060 | 108.771        | 0.024       | 78.54%         |
| 1.0065 | 108.816        | 0.016       | 84.21%         |
| 1.0070 | 108.873        | 0.0025      | 89.34%         |

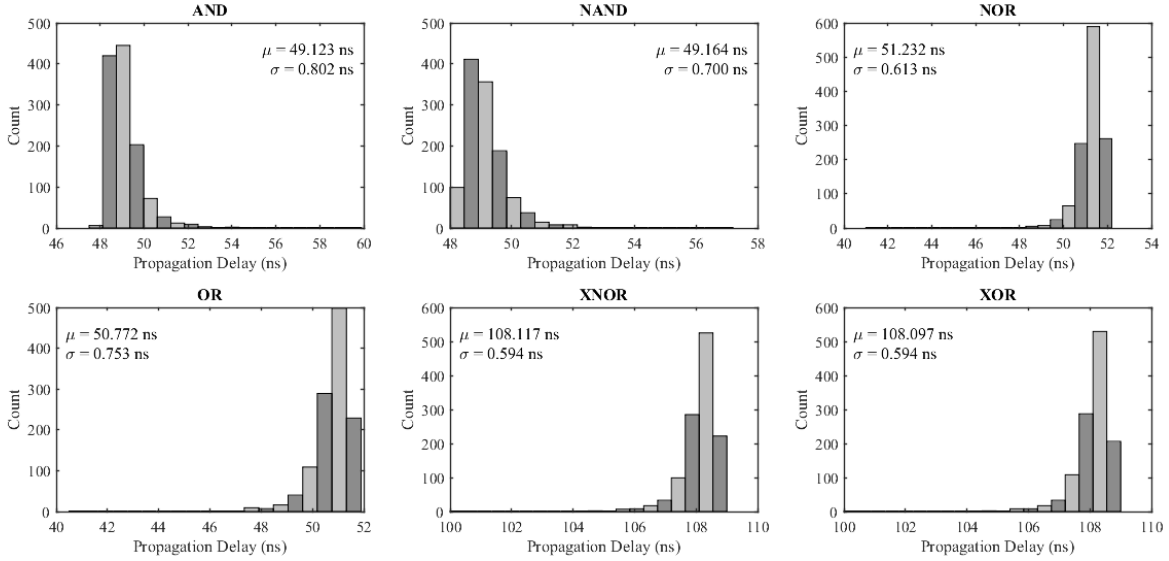


Fig. 4: Propagation delay distributions obtained from 1200-point Monte Carlo simulations for all Boolean logic gates implemented in the proposed CIM SRAM architecture.

testing batch size = 256, dataset = MNIST, activation = binary sign activation, pooling = MaxPool2D ( $2 \times 2$ ), convolution layers = 2 (32 and 64 filters with  $3 \times 3$  kernels), fully connected layer =  $3136 \rightarrow 10$ , Monte Carlo runs = 10, clock margin sweep = 1.005 to 1.007 with 20 points. The architecture of the BNN model is shown in Fig. 5. Next, the MNIST handwritten digit dataset is used to evaluate the classification accuracy across different timing-induced XNOR errors. The neural network model is trained for 10 epochs to obtain a stable baseline accuracy before error injection. The clock margin range of 1.005–1.007 was chosen to analyze the near-threshold operating region where timing violations transition from frequent to rare. This range captures the critical trade-off between high-speed operation and BNN inference reliability while avoiding excessively unstable or fully ideal operating conditions. Table I summarizes the relationship between clock margin, timing error probability, and MNIST classification accuracy. As the clock margin increases, the timing-induced XNOR error probability decreases, resulting in improved BNN inference accuracy. In the evaluated range  $k = 1.005 - 1.007$  reducing the XNOR error probability from approximately 9.5% to 0.25% improves the MNIST classification accuracy from 58.27% to 89.34% as seen in Fig. 6 & 7.

This study demonstrate the strong dependence of application-level accuracy on circuit-level timing reliability. Small probability of timing errors in the fundamental XNOR logic operations can substantially degrade the accuracy of neural network inference. The obtained results highlight the importance of cross-layer analysis when designing SRAM-based compute-in-memory systems for machine learning workloads. By correlating circuit-level timing variability to neural network accuracy, designers can determine appropriate operating points that balance performance, reliability, and

energy efficiency.

## VI. SYSTEM-LEVEL EVALUATION

### A. Inference Robustness under Timing Errors

Monte Carlo inference simulations are used to evaluate the influence of circuit-level timing variability on system-level performance, utilizing the extracted XNOR timing error model. The selected operating point corresponds to a clock margin factor of  $k = 1.00625$ , which results in an XNOR error probability of  $P_{error} = 0.016$ . BNN inference is repeatedly executed under the modeled error condition, where XNOR operations are probabilistically corrupted based on the delay-derived circuit error model. The resulting inference accuracy distribution is shown in Fig. 8. The narrow spread indicates stable performance despite stochastic XNOR errors, confirming that BNNs are robust to small logic-level computation errors, consistent with prior findings in compute-in-memory accelerators. [7], [12].

### B. Energy Consumption Analysis

A Monte Carlo energy analysis is performed in Cadence using 1200 random samples to evaluate the energy behavior of the proposed CIM architecture under process variations. The energy of the IMC logic path is first extracted at the circuit level and then propagated to system-level inference energy based on the number of Boolean operations required by the BNN. The inference energy is estimated using

$$E_{inf} = N_{ops} \times E_{op} \quad (4)$$

where  $N_{ops}$  denotes the total number of Boolean operations per inference and  $E_{op}$  represents the energy per CIM operation obtained from circuit simulations. For the implemented BNN architecture, the convolutional and fully connected layers

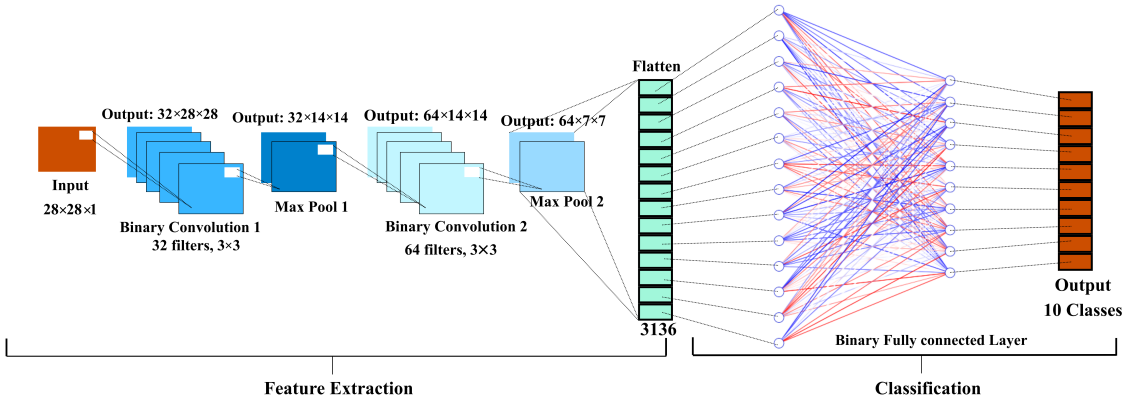


Fig. 5: Binary neural network architecture used for hardware-aware evaluation. Two binary convolution layers with max-pooling extract features from  $28 \times 28 \times 1$  MNIST images. The resulting feature maps are flattened and classified by a binary fully connected layer into 10 classes using XNOR-based accumulation.

require approximately  $3.87 \times 10^6$  Boolean operations per inference.

The resulting inference energy distribution is shown in Fig. 9. The mean inference energy is approximately  $12.27 \mu\text{J}$  with a standard deviation of  $0.995 \mu\text{J}$ . Across the Monte Carlo samples, the inference energy ranges from  $9.33 \mu\text{J}$  to  $15.18 \mu\text{J}$ , indicating that the proposed SRAM-based CIM architecture maintains stable energy characteristics under process variations.

## VII. COMPARISON WITH PRIOR WORK

SRAM-based compute-in-memory (CIM) architectures reduce data movement and mitigate the von Neumann bottleneck. Prior work demonstrated in-memory Boolean logic in SRAM arrays [1] and digital CIM macros supporting neural-network workloads [5], [9], [10]. Standard 6T SRAM cells have also been adapted for in-memory Boolean and arithmetic operations [2]. SRAM-CIM has been explored for neural-network acceleration, including reconfigurable and approximate CIM designs [5], [7], as well as accelerators for spiking and application-specific neural networks [3], [6]. While prior

TABLE II: Comparison with Prior SRAM-CIM Works

| Work                     | Boolean CIM | Variation / Nonideality | Cross-Layer Evaluation |
|--------------------------|-------------|-------------------------|------------------------|
| X-SRAM [1]               | Yes         | PVT, $V_T$ mismatch     | No                     |
| Dhakad <i>et al.</i> [2] | Yes         | Not reported            | No                     |
| Wang <i>et al.</i> [9]   | Partial     | Not reported            | No                     |
| Lee <i>et al.</i> [10]   | Partial     | Voltage range           | No                     |
| Kim <i>et al.</i> [7]    | No          | Not reported            | Yes                    |
| ASIM [12]                | No          | ADC quant., noise       | Yes                    |
| <b>This Work</b>         | <b>Yes</b>  | <b>Timing variation</b> | <b>Yes</b>             |

studies highlight the impact of circuit-level nonidealities [12], most focus on macro-level metrics such as throughput and energy efficiency [5], [9], [10]. In contrast, this work links timing-induced XNOR errors from Cadence delay simulations to BNN inference accuracy. Exploring alternative SRAM technologies such as FinFET and CNTFET cells [13]–[17], along with variability-aware SRAM studies remains an important direction for future reliability-aware CIM analysis.

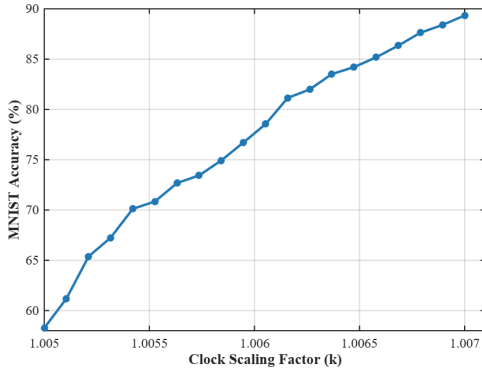


Fig. 6: Impact of timing-induced XNOR errors on MNIST classification accuracy as a function of clock margin.

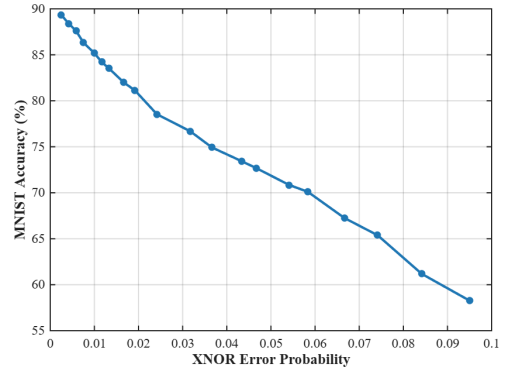


Fig. 7: MNIST classification accuracy as a function of timing-induced XNOR error probability.

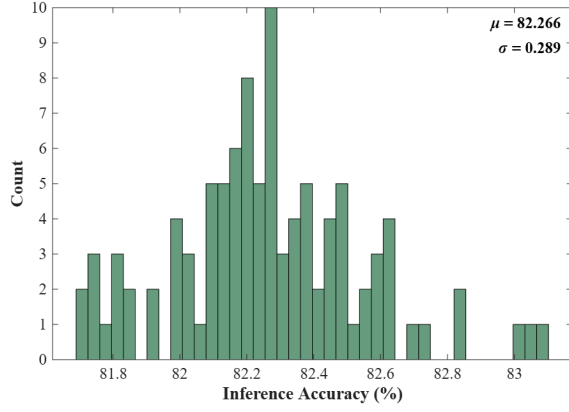


Fig. 8: Distribution of MNIST inference accuracy at the selected operating point ( $k = 1.00625$ ,  $P_{error} = 0.016$ .)

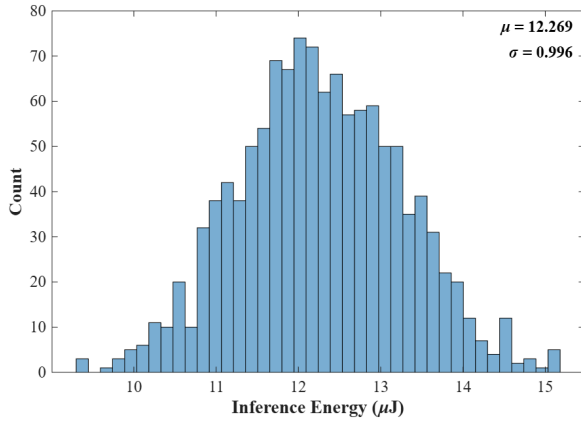


Fig. 9: Distribution of the estimated BNN inference energy derived from circuit-level CIM energy simulations under process variations.

## VIII. CONCLUSION

This work presents a 6T CMOS SRAM-based compute-in-memory (CIM) architecture capable of performing multiple Boolean logic operations directly within memory arrays. The proposed design enables in-memory evaluation of logic functions. The circuit is implemented and verified using transistor-level simulations in the Cadence Virtuoso environment with a 45nm CMOS technology library. Monte Carlo simulations are performed to analyze the impact of process variations on propagation delay for each Boolean logic gate. Using 1200 Monte Carlo samples, the delay distribution of the XNOR operation is characterized, yielding a mean delay of 108.183 ns and a standard deviation of 488.07 ps. To understand the system-level implications of these circuit-level variations, the extracted XNOR error probabilities were injected into a hardware-aware Binary Neural Network (BNN) model for MNIST digit classification. The results demonstrate a clear relationship between timing reliability and neural network accuracy, where reducing the XNOR error probability from approximately 9.5% to 0.25% improves inference accuracy from

about 58.27% to 89.34%. This work puts forward a circuit-to-application evaluation approach that correlates circuit-level timing variation of SRAM-based Boolean CIM operations with the inference accuracy of binary neural networks (BNNs). The proposed methodology provides useful insights for designing reliable SRAM-based CIM systems for neural-network workloads.

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